

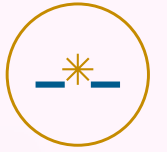


Seasonal adjustment of econometric time series

EVOLUTION | THEORY | PRACTICE

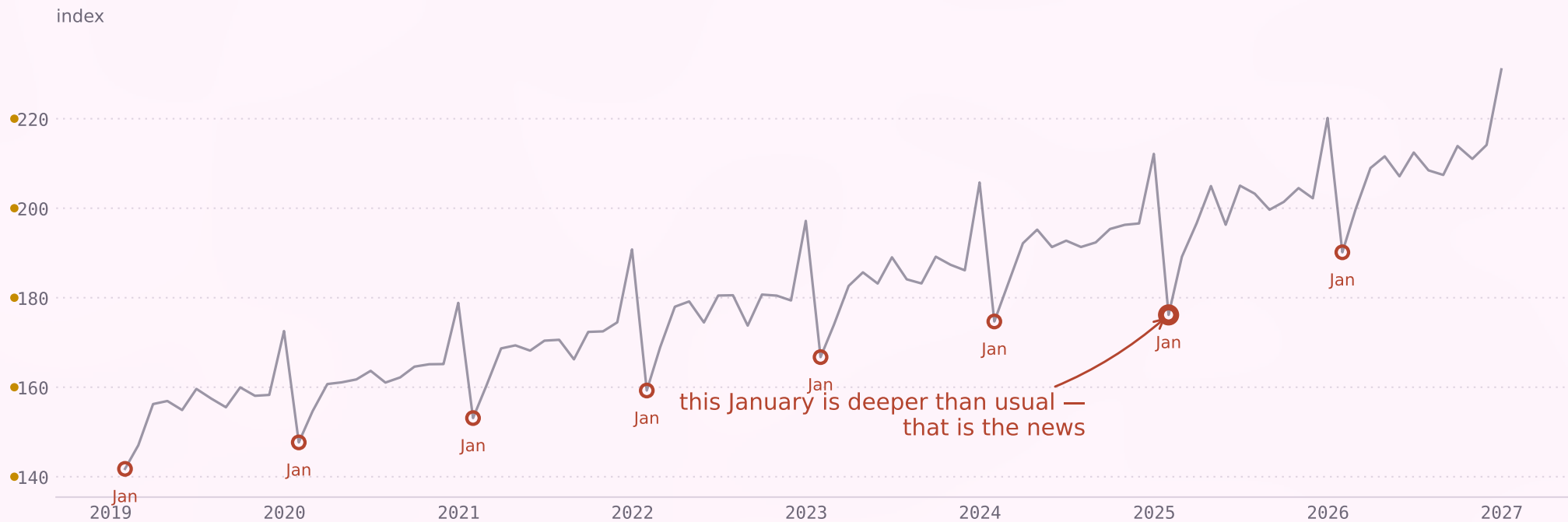
exit10.ru

github.com/MikeMike13/SEASONAL



The series falls every January — and that means nothing

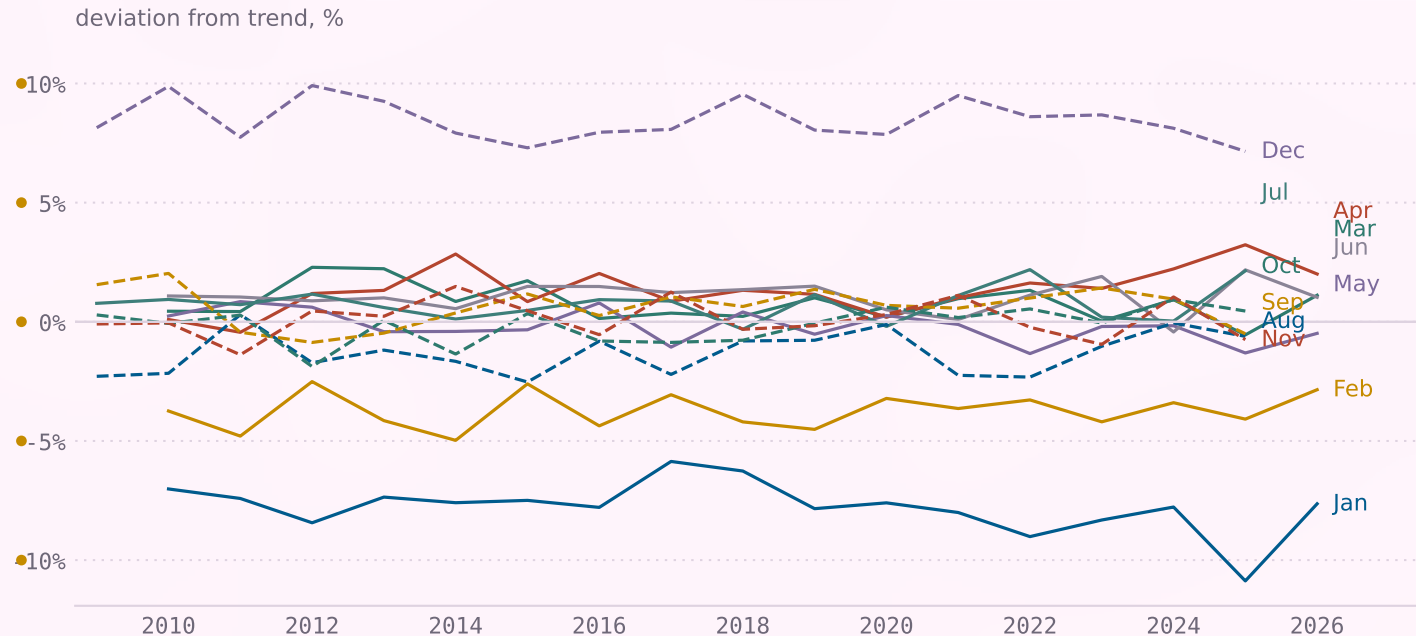
what matters is only whether the fall is deeper than usual



DEFINITION

Seasonality is a profile of the year that repeats every year

each line is one calendar month; years run along the horizontal



December is steadily above the trend, January below. The distance between the lines is the seasonal swing.

Lines converging to zero — seasonality is weakening.

A line stepping to another level — the profile of that month has broken.

One metric will fool you, three at once almost never will

all three run on a detrended series, otherwise the trend drowns the signal

metric	what it catches	sign of success
F-test on seasonal dummies	is there a wave at all	$p > 0,05$
autocorrelation, lag 12	repeatability a year apart	inside the band
share of seasonal power	energy at the seasonal frequencies	halved or more

Next: what the second and third metrics look like before and after adjustment.

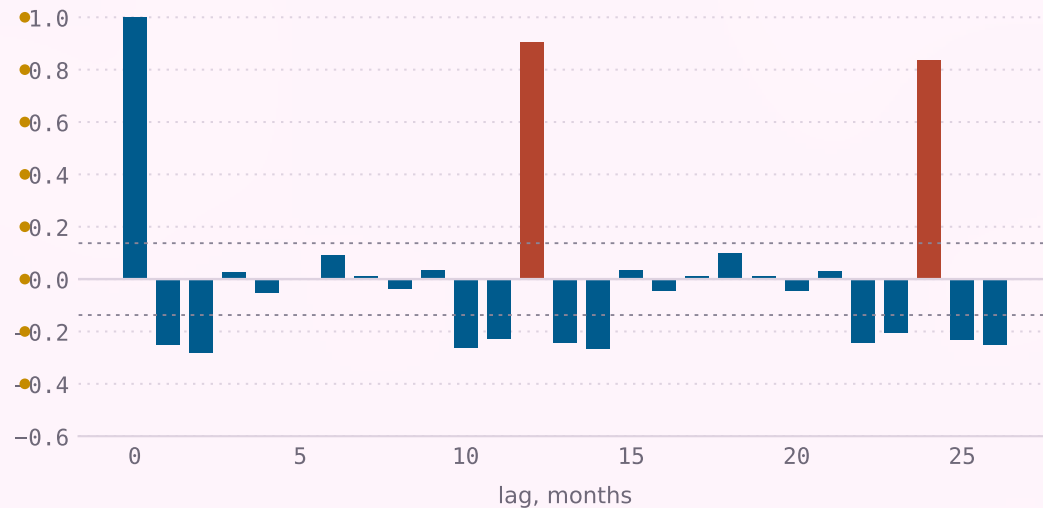
Autocorrelation: before and after

the bar at lag k shows how alike values k months apart are

The one on the left is the series against itself. The bar at 12 shows how much December resembles last December: that is seasonality.

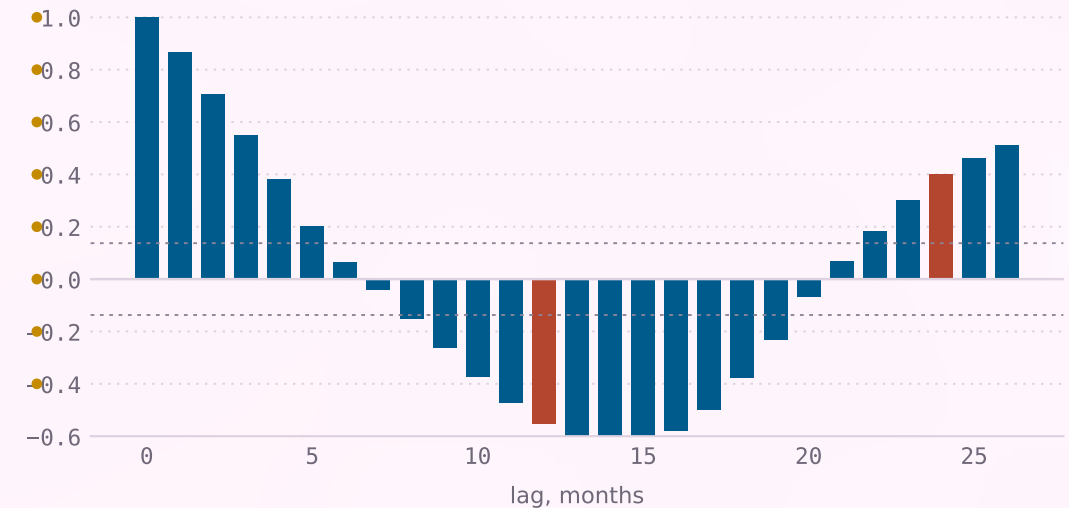
Before adjustment

ACF(12) = +0.90 — far beyond the band



After adjustment

ACF(12) = -0.55 — inside the band

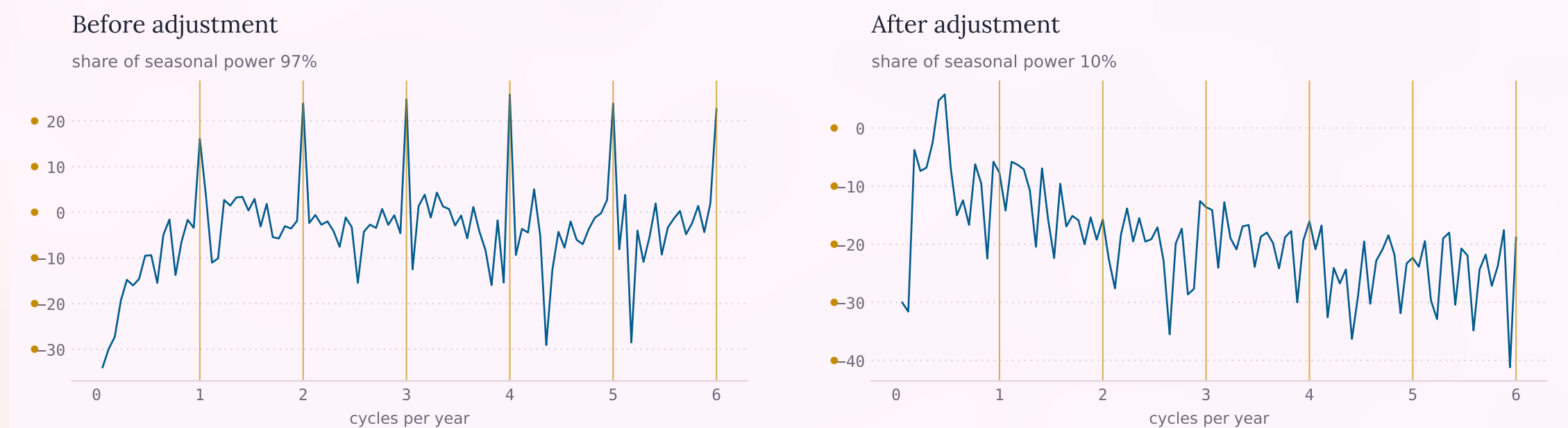


Coral bars are the seasonal lags. The tall bar at lag 1 survives the adjustment: that is inertia, not seasonality.

Periodogram: where the energy of the seasonal frequencies went

the series is decomposed into waves; the chart shows which wavelengths dominate

A peak at one is a wave exactly a year long, at two a half-year wave, then shorter harmonics. That is seasonality.



The gold lines are the seasonal frequencies. On the right there are no peaks: that energy is gone.



I

The decisions made by the analyst

the program decides none of them

The whole process on one page

one slide per decision from here on

01	Is there seasonality	F-test, ACF, spectrum	if not — do not adjust
02	Level or rates	convert to a level	rates break the typing of outliers
03	The logarithm	the swing-level test	swing grows with the level — take logs
04	The calendar	working days, Easter	a flow and a stock are tested differently
05	Outliers	AO, LS, TC and SO separately	candidates from the algorithm, the decision is yours
06	The method	comparison by revisions	not by the smoothness of the result

Every decision is recorded with its reason: a year from now that is the only way to know why.

Adjust levels, not rates

rates always follow from an adjusted level; the reverse does not hold

- ◆ in a series of rates a level shift is indistinguishable from a one-off spike
- ◆ the typing of outliers breaks: an LS looks like an AO
- ◆ a year-to-date total is first decumulated into months
- ◆ indices with a moving base are rebased to a single base

The level first, any derived measures afterwards

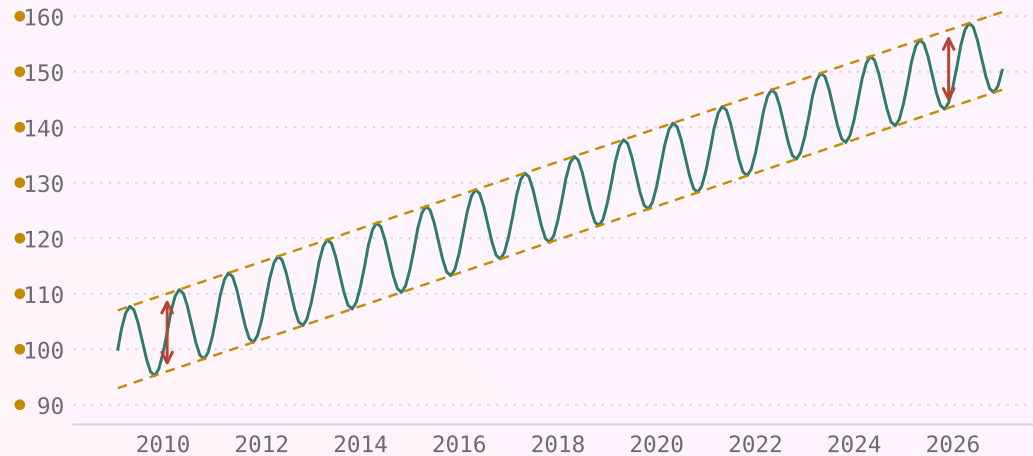
A logarithm if the swing grows with the level

forgetting it is more dangerous than it seems: the error builds up at the right edge

The swing-versus-level test: the series is cut into years and the swing inside a year is compared with the level of that year. A slope near 1 calls for a logarithm, near 0 does not.

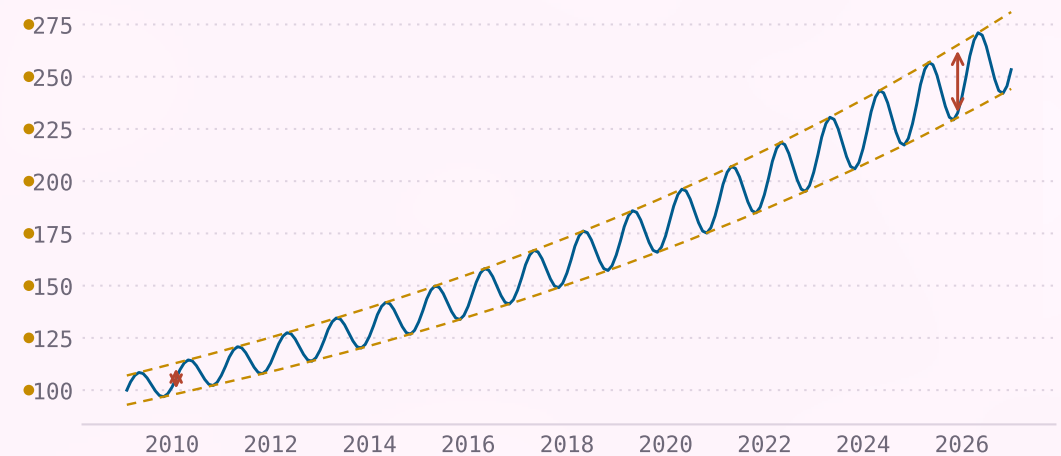
Additive

the swing is the same: 7 units at both ends



Multiplicative

the swing grows: the same 7% — but of a larger level



The gold dashes are the envelopes of the swing; the coral arrows measure it at both ends.

The calendar: working days and Easter are different effects

and they work differently for a flow and for a stock

indicator type	examples	what matters	regressor
flow	output, retail, freight	how many working days are in the month	td
stock	money supply, reserves	the weekday of the reporting date	tdstock[31]

31 means not “the thirty-first” but “the last day of the month”.

Easter does not change the number of working days — it shifts demand between months.
Russian series need the Orthodox date: with the Western one the regressor barely correlates with the real effect.

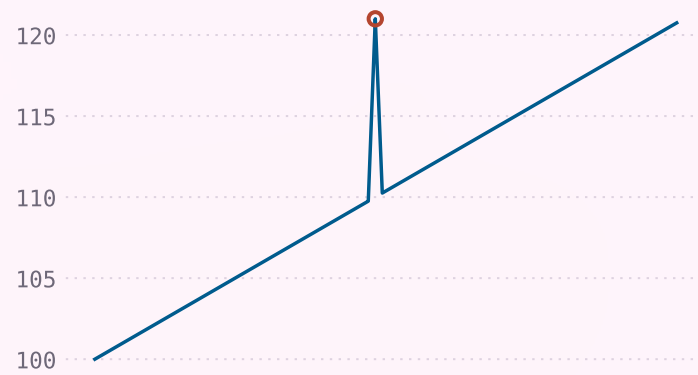
For a flow significance is a t-statistic; for a stock a joint F-test over six weekday columns.

Three types of outliers, three different logics

if they are not removed during estimation they leak into the seasonality and the trend

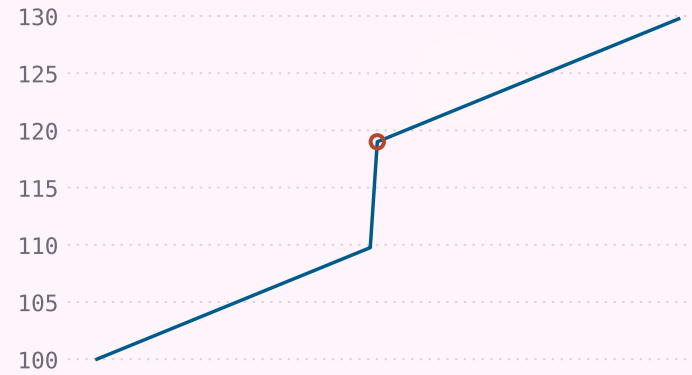
AO

a one-off spike, next month as before



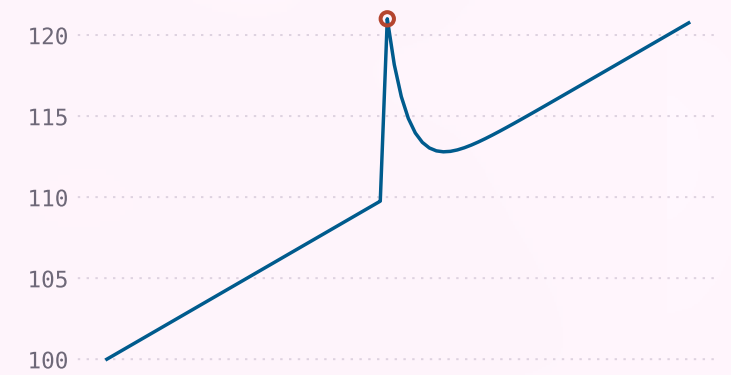
LS

the level has shifted for good



TC

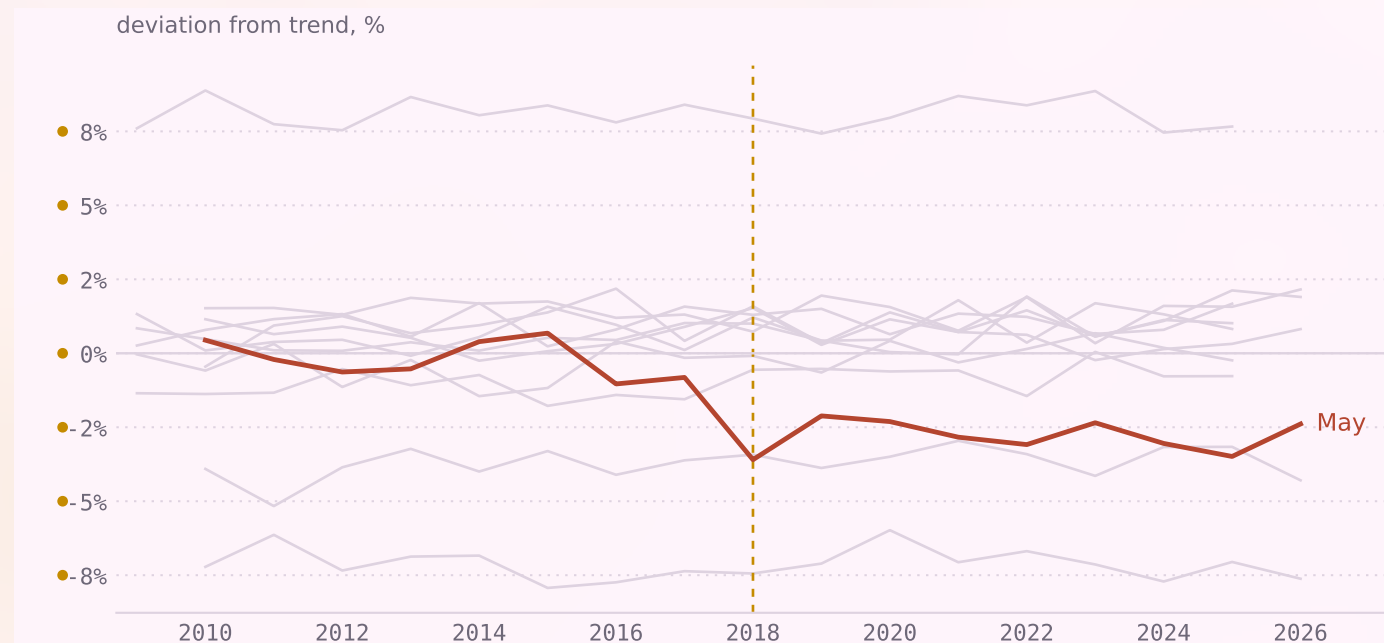
a spike fading over months



An outlier is not erased: its estimated effect is subtracted and then restored in the final series — the 2020 slump is real, not an artefact of the adjustment.

The seasonal outlier SO: the profile of one month has broken

the level of the series has not shifted — the distribution within the year has



For years May was above the trend; from 2018 it moved below and stayed there.

Neither AO nor LS nor TC describes this: an AO returns, an LS shifts the whole series.

X-13 does NOT search for seasonal outliers — its search takes only AO, LS, TC.

What we do: spot it by eye on the month sub-series and set it by hand as SO2018.5 in the specification.



II

ARIMA: the language for describing a series

about decomposition only; how it is used comes in the next section

Wold's theorem: why a series can be decomposed at all

Herman Wold, 1938 — the result from which linear time-series econometrics grew

any stationary series = a predictable part + an infinite moving average of shocks

- ◆ the “predictable part” is what follows from the past: trend, cycle, seasonality
- ◆ the “shocks” are what is new: weather, decisions, crises
- ◆ a shock does not act only today — it fades, and the fading is described by weights
- ◆ so a series can be written with finitely many parameters instead of its whole history

ARIMA is a practical shorthand for that result: AR and MA approximate the infinite sum compactly, and I brings a non-stationary series to the form where the theorem applies.

Three letters and what stands behind each

the notation ARIMA(p, d, q): how many past values, differences and past shocks to take

$$\Delta^d y(t) = c + \underbrace{\varphi_1 \cdot \Delta^d y(t-1) + \dots + \varphi_p \cdot \Delta^d y(t-p)}_{\text{AR: } p \text{ past values}} + \underbrace{\varepsilon(t) + \theta_1 \cdot \varepsilon(t-1) + \dots + \theta_q \cdot \varepsilon(t-q)}_{\text{MA: } q \text{ past shocks}}$$

I: d differences

AR: p past values

MA: q past shocks

	name	the point in one phrase	parameter
AR	autoregression	today depends on yesterday's value	p
I	integration	we work with changes, not levels	d
MA	moving average	today remembers yesterday's shock	q

d is usually 0 or 1: one difference is enough to remove a trend.
More than two is almost always a misspecification, not a property of the series.

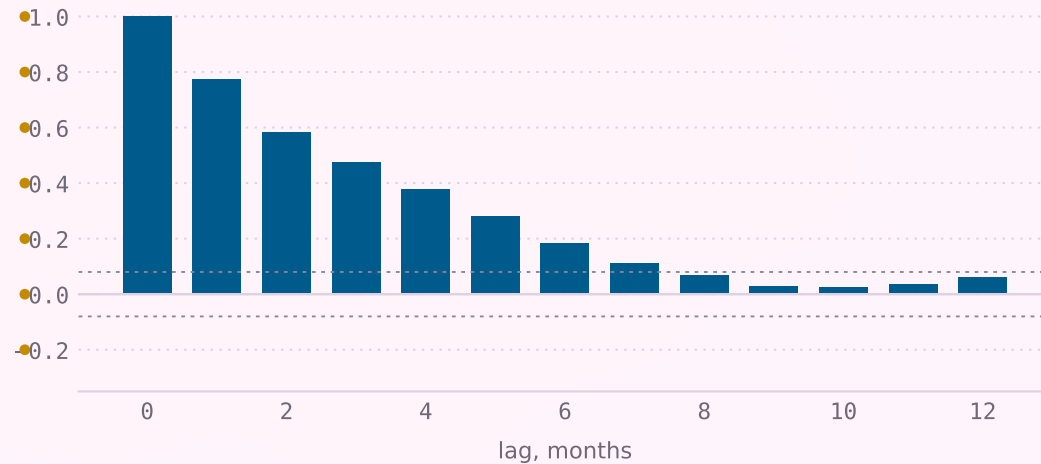
AR or MA: each leaves its signature on the autocorrelation

for MA(1) the autocorrelation stops after the first lag; for AR(1) it decays smoothly

The order is chosen automatically by AICC, but the ACF is still worth a look: it shows what exactly the model is trying to describe.

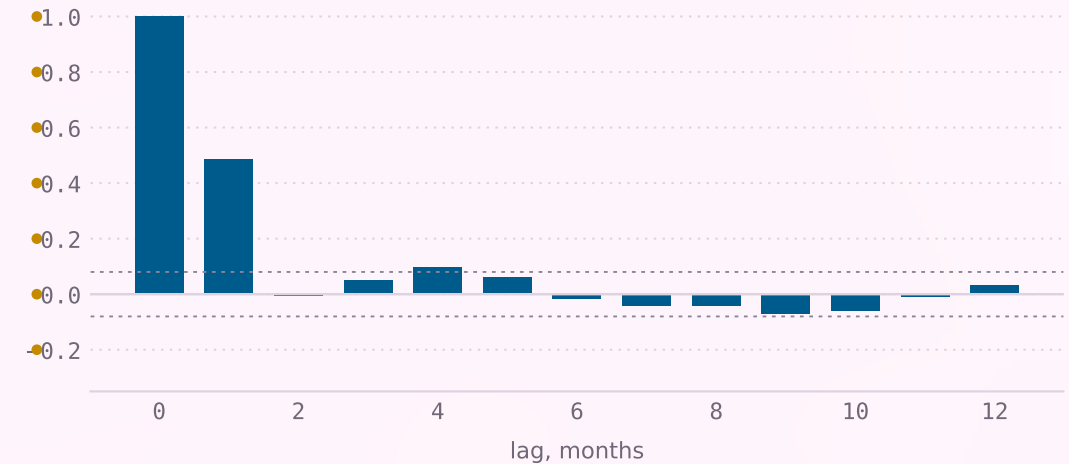
AR(1), coefficient 0.75

memory lingers: each lag weaker than the last



MA(1), coefficient 0.8

memory of exactly one month: zeros beyond



The band around zero is the noise limit: bars inside it are indistinguishable from zero.

The seasonal part: the same, but a year apart

the second bracket describes the link to the same month a year ago

$$\text{ARIMA}(p, d, q)(P, D, Q)_s$$

$(0\ 1\ 1)(0\ 1\ 1)_{12}$ — the “airline model”, where the automatic search lands for most monthly series

- ◆ $D = 1$ — seasonal differencing: last May is subtracted from this May
- ◆ $s = 12$ for months, 4 for quarters
- ◆ models are comparable by AICC only at equal d and D : otherwise the data differ
- ◆ orders above $(1, 1, 1)$ are rare on monthly series

regARIMA: the regression and the ARIMA are estimated together

the calendar and the outliers are not pre-processing but part of one model

$$y(t) = \beta \cdot x(t) + z(t), \quad z(t) \sim \text{ARIMA}(p, d, q)(P, D, Q)_s$$

- ◆ $x(t)$ — working days, Easter, AO/LS/TC/SO outliers, your own regressors
- ◆ estimation is joint: otherwise the calendar effect leaks into the ARIMA and a shock into the calendar
- ◆ outliers distort the seasonal estimate, which is why they are removed here
- ◆ and above all — the model can extend the series forwards and backwards

The last point is why the X-11 filter needed a model at all:
extending the series by a forecast closes the edge problem. That is the next section.



III

The evolution of methods: what each step solved

1954 → 1965 → 1980 → 1998 → 2012

Census Method I and II: averages by calendar month

the first machine method — a UNIVAC program, hence the simplicity

- ◆ a month's seasonal factor is its average deviation over all years
- ◆ the weights are equal: 1954 counts as much as last year
- ◆ what it solved: it removed hand-tuning and made the adjustment reproducible
- ◆ what it could not do: it treats seasonality as fixed, and it is not

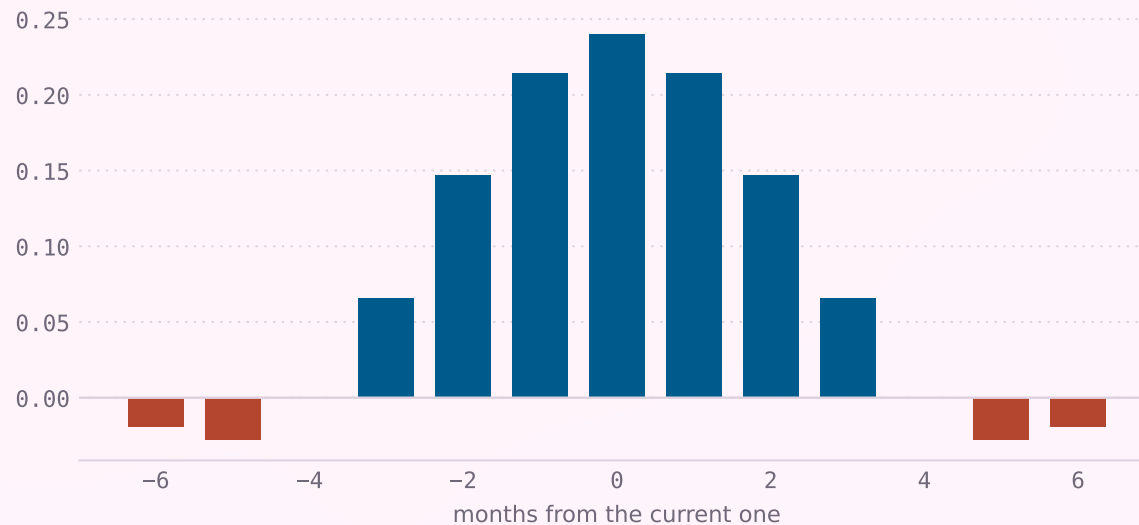
On a series whose swing doubled over twenty years the method gives an average profile: too much seasonality at the start, too little at the end.

X-11: moving filters instead of averages

seasonality is estimated locally, so it may change over time

Weights of the 13-term Henderson filter

nearer months carry more weight than distant ones



What it solved:
seasonality became movable.

How: instead of an average over all years — 3×3 and 3×5 moving averages over the same month, and a Henderson filter for the trend.

The negative weights at the ends are no mistake: that is how the filter keeps the slope of the trend instead of flattening it.

What it could not do: at the last month there is no symmetric filter — and the estimate is heavily revised.

X-11-ARIMA: extend the series to fix the edge

Statistics Canada, Dagum and Hillmer — the answer to the revisions problem

- ◆ the series is extended by an ARIMA forecast a year forwards and backwards
- ◆ the ordinary symmetric filter is applied to the extended series
- ◆ what it solves: the edge is computed like the middle
- ◆ the effect: revisions fall by roughly half

The point: the forecast is not wanted for itself but as a prop for the filter.
Its quality affects only the edge of the series — and that is where the value is.

X-12-ARIMA and X-13ARIMA-SEATS

first the model absorbed the calendar and the outliers, then a second method joined the filter

version	what it added	why it matters
X-12-ARIMA, 1998	regARIMA inside	the calendar and the outliers are estimated together with the model
	automatic order selection	the specification stopped being manual
X-13ARIMA-SEATS, 2012	the SEATS module	a second way of decomposing in the same program
	the M and Q diagnostics	formal quality criteria for the adjustment

X-13 is not a new algorithm but a union: the same X-11 inside, the same regARIMA, plus SEATS as an alternative branch at the last step.

SEATS: the filter is not set, it is derived from the model

Signal Extraction in ARIMA Time Series — Gómez and Maravall, Bank of Spain

- ◆ once the ARIMA is chosen, the variance structure of the series is known
- ◆ from it follows a decomposition into components, each with its own ARIMA
- ◆ the filter comes out automatically and fits this very series
- ◆ and a confidence interval as a bonus: X-11 has none at all

The price: the result depends entirely on whether the model is right.
X-11 is more robust to a wrong model — its filter is fixed in advance.

So the X-11 versus SEATS argument is unsettled: US offices lean to X-11,
European ones to SEATS.



IV

Beyond X-13: who solves what

TRAMO and JDemetra+: the same idea, a different school

TRAMO — Time series Regression with ARIMA noise, Missing values and Outliers

- ◆ TRAMO does what regARIMA does: calendar, outliers, missing values, the model
- ◆ the differences are in selection thresholds and in how the logarithm is decided
- ◆ historically TRAMO was written for SEATS — both come from the Bank of Spain
- ◆ JDemetra+ — a Java implementation, recommended by Eurostat since 2015

	X-13	JDemetra+	
regARIMA + X-11	yes	yes	Pre-processing and decomposition are independent choices.
regARIMA + SEATS	yes	yes	
TRAMO + X-11	—	yes	Two combinations exist only in JDemetra+.
TRAMO + SEATS	—	yes	

When the official method is not enough

all of these solve problems X-13 was never built for

method	what it adds beyond X-13	when to use it
STL	robustness to outliers	dirty data, no time to mark it up
MSTL	several seasonalities at once	daily series: the week and the year together
TBATS	fractional period, long cycles	365.25 days, complex periodicity
Prophet	many holidays and breaks	a quick result with no tuning
UCM	an honest interval for the trend	you need uncertainty, not a point
wavelets	WHEN the series changed character	finding a structural break

None of them is official: statistical offices publish X-13 or JDemetra+.
The rest are tools for analysis, not for publication.

3MMA SAAR: the growth rate right now

a three-month average of the adjusted series, recast as an annual rate



Year on year depends on the base: last year's slump by itself creates a high rate today.
SAAR looks only at the recent months and answers a different question — how fast are we going now.

The order of work on a real series

- 01 check whether there is seasonality — if not, do not adjust
- 02 convert to a level; the logarithm by the swing-level test
- 03 the calendar: flow or stock, Easter separately
- 04 outliers: candidates from the algorithm, the decision from the analyst
- 05 choose the method by the size of revisions, not by smoothness
- 06 record the decisions with reasons and keep them beside the calculation

The edge is always unreliable: the last months will be revised as new data arrive.

All of this can be run on your own machine

notebooks, module and data — an open repository

file

what for

x13_seasonal_full_pgh.ipynb

the full walkthrough, ~160 cells with checks on data

x13_seasonal_short_pgh.ipynb

an hour-long overview

indicator_template_pgh.ipynb

a template for your own series

seasonal_toolbox.py

the module, ~160 functions

The X-13ARIMA-SEATS binary is downloaded separately, from the US Census Bureau:

[census.gov/data/software/x13as.html](https://www.census.gov/data/software/x13as.html)

github.com/MikeMike13/SEASONAL



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